

# Servas Adolph Tarimo

PhD Researcher | Future Convergence Technology / Big Data Engineering  
Advanced Data Mining Lab (ADM Lab), Soonchunhyang University, South Korea

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## PERSONAL INFORMATION

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|----------------------|--------------------------------------------------------------|
| <b>Nationality</b>   | Tanzanian                                                    |
| <b>Languages</b>     | Swahili (Native), English (Professional working proficiency) |
| <b>Research base</b> | Soonchunhyang University, Asan, South Korea                  |

## PROFESSIONAL SUMMARY

Servas Adolph Tarimo is a PhD researcher in Future Convergence Technology / Big Data Engineering at Soonchunhyang University, South Korea. His work focuses on AI for healthcare, especially medical image analysis, domain adaptation, multimodal clinical report generation, and practical LLM/RAG systems. He develops AI models that remain reliable across hospitals, imaging devices, clinical environments, and patient populations where data distributions differ. His long-term goal is to build practical, trustworthy AI health systems for real-world settings, including Tanzania and other low-resource contexts.

## SKILLS

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>               | Python, Java, C++, JavaScript, React PWA, FastAPI                                                                 |
| <b>Machine Learning &amp; AI</b> | PyTorch, TensorFlow, Keras, Scikit-Learn, YOLO, Vision Transformers, GANs                                         |
| <b>LLM / AI Systems</b>          | Retrieval-Augmented Generation (RAG), LangChain/LangGraph-style workflows, NLP, clinical decision-support systems |
| <b>Data Analysis</b>             | Healthcare data analytics, statistical modeling, multimodal data analysis, hybrid search, EHR data analysis       |
| <b>Tools</b>                     | Jupyter Notebook, VS Code, Git, GitHub, OpenCV, Supabase/PostgreSQL                                               |
| <b>Healthcare AI</b>             | White blood cell classification, medical image segmentation, domain adaptation, clinical report generation        |

## EDUCATION

|                                                                                                                                                                     |                                                                                                                                                                                                       |                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 2023.09 - Present                                                                                                                                                   | <b>Soonchunhyang University</b><br><b>Ph.D. Student, Future Convergence Technology / Big Data Engineering</b><br><b>Advisor: Prof. Woo Ji-Young</b><br><b>Lab: Advanced Data Mining Lab (ADM Lab)</b> | Asan, South Korea       |
| 2021.09 - 2023.08                                                                                                                                                   | <b>Soonchunhyang University</b><br><b>M.Sc. in Big Data Engineering</b><br><b>Advisor: Prof. Woo Ji-Young</b><br><b>Lab: Advanced Data Mining Lab (ADM Lab)</b>                                       | Asan, South Korea       |
| * Dissertation: WBC YOLO-ViT: 2-Way 2-Stage White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer. Paper   View thesis. |                                                                                                                                                                                                       |                         |
| 2017.01 - 2020.08                                                                                                                                                   | <b>The United African University of Tanzania (UAUT)</b><br><b>B.Sc. in Computer Engineering and Information Technology</b>                                                                            | Dar es Salaam, Tanzania |

## RESEARCH EXPERIENCE

2023 - Present

**Soonchunhyang University**  
**Department of Future Convergence Technology / Big Data Engineering**  
**Advisor: Prof. Woo Ji-Young**

Asan, South Korea

- Domain Adaptation in Medical AI - developing transfer learning and domain adaptation methods for generalizing medical imaging and healthcare models across clinical sites, imaging protocols, and patient populations.
- Multimodal Report Generation for Blood Smear Analysis - building AI systems that read microscopy images from blood smear tests and generate structured clinical reports by combining image data, attention mechanisms, and language models.
- Medical Report Generation & RAG for Clinical Decision Support - building LLM-based systems that produce structured clinical reports and physician-style patient summaries grounded in verified medical literature via Retrieval-Augmented Generation.

2021 - 2024

**Soonchunhyang University**  
**Department of Future Convergence Technology / Big Data Engineering**  
**Advisor: Prof. Woo Ji-Young**

Asan, South Korea

- Blood Cell Classification & Domain Adaptation - developed methods for classifying white blood cells and other blood cells across hospitals with different equipment, staining conditions, and image distributions.
- Weight Module for cross-hospital adaptation - designed model-weighting strategies to reduce misleading feature learning when source and target hospital domains do not fully match.
- Reinforcement learning-based style transfer - investigated approaches for bridging visual differences between hospitals and improving robustness in blood-cell image analysis.
- Generative Adversarial Networks for Medical Data - developed GAN-based methods for generating realistic blood-cell images to augment limited medical datasets and support model training.

## RESEARCH INTEREST

- **Domain Adaptation in Medical AI:** developing AI models that keep working when transferred across hospitals, imaging devices, clinical protocols, and patient populations.
- **Blood Cell Image Analysis:** automated detection, counting, classification, and report generation for blood-smear microscopy and hematological diagnostics.
- **Multimodal Clinical AI:** combining medical images, clinical notes, and structured patient data to support richer clinical understanding.
- **LLM and RAG Systems:** retrieval-grounded AI applications for clinical decision support and practical user-facing guidance systems.
- **AI for Low-Resource Settings:** building practical AI tools for healthcare and education contexts where access to specialists or expert guidance is limited.

## WORK EXPERIENCE

2021.09 - Present      **Research Assistant**      Asan, South Korea  
**Advanced Data Mining Lab (ADM Lab), Soonchunhyang**  
**University**

- Developing medical AI systems for blood-cell image analysis, domain adaptation across clinical sites, multimodal report generation, and RAG-based clinical decision-support workflows with Prof. Woo Ji-Young.

## PARTICIPATING PROJECTS

|                                  |                                                                                                                                        |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>2026 - Present</b>            | Domain Adaptation in Medical AI - robust healthcare models across hospitals, imaging devices, and patient populations.                 |
| <b>2026 - Present</b>            | Multimodal Report Generation for Blood Smear Analysis - microscopy-to-report generation for clinical workflows.                        |
| <b>2026, Completed</b>           | Matokeo Yangu - bilingual platform for Tanzanian students to check exam results and receive AI-guided university and career advice.    |
| <b>2021.09 - 2024, Completed</b> | WBC Detection & Classification - automated white blood cell detection, counting, and classification with YOLO and Vision Transformers. |
| <b>2021 - 2024, Completed</b>    | Generative models for blood-cell image augmentation and class-imbalance mitigation.                                                    |

## INTERNATIONAL JOURNAL ARTICLES [\*: CO-FIRST OR CORRESPONDING AUTHOR]

1. Yunjung Hong, Servas Adolph Tarimo, and Jiyoung Woo, "Pancreas Segmentation Using a Two-Stage Pipeline of Faster R-CNN and TransUNet," *Applied Sciences*, vol. 16, no. 12, article 5764, 2026. doi:10.3390/app16125764.
2. S.A. Tarimo, Mi-Ae Jang, Emmanuel Edward Ngasa, Hee Bong Shin, HyoJin Shin, and Jiyoung Woo, "WBC YOLO-ViT: 2-Way 2-Stage White Blood Cell Detection and Classification with a Combination of YOLOv5 and Vision Transformer," *Computers in Biology and Medicine*, vol. 169, article 107875, 2024. doi:10.1016/j.combiomed.2023.107875.
3. Emmanuel Edward Ngasa, Mi-Ae Jang, S.A. Tarimo, Jiyoung Woo, and Hee Bong Shin, "Diffusion-based Wasserstein Generative Adversarial Network for Blood Cell Image Augmentation," *Engineering Applications of Artificial Intelligence*, vol. 133, article 108221, 2024. doi:10.1016/j.engappai.2024.108221.

## INTERNATIONAL CONFERENCE

1. S.A. Tarimo, "Adapting YOLO-ViT for Differential Diagnosis of Myelodysplastic Syndromes and Normal Blood Cell," *Proceedings of the Korea Society of Computer and Information Conference*, 2024.
2. S.A. Tarimo and J. Woo, "White Blood Cell Detection and Classification using YOLOv5 with Hybrid ResNet50-VGG16-SVM," in *Proceedings of the 6th International Conference on ICT for Smart Health & Home (ICT4sHealth & Home)*, Kota Kinabalu, Malaysia, December 18-22, 2022. Presentation: December 19, 2022.

## **PATENTS**

1. None. Open to future innovations and patent opportunities related to AI, healthcare, clinical decision support, and medical image analysis.

## AWARDS AND HONORS

- 2010 Excellence Award - Certificate of Computer Maintenance and Repair, Microsoft Office 2003 and 2007.
- 2010 Excellence Award - Certificate of Network Setup, Internet Repair, Email Usage, and Web Design.
- 2012 Excellence Award - Certificate of IT Essentials: PC Hardware and Software, Cisco Networking Academy.
- 2014 Excellence Award - Certificate of IT Essentials: PC Hardware and Software, Cisco Networking Academy.
- 2019 Excellence Award - Certificate Software Developer Program and Adobe Photoshop Design.
- 2019 Excellence Award - Certificate of Korea Culture Exchange Program (PASS Program).

## REFEREE

Prof. Woo Ji-Young, Ph.D.  
Department of AI and Big Data Engineering, Soonchunhyang University  
Asan, South Korea  
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